

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	18 june 2026
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Standard Form (Username, Email, Password) • Registration through Gmail integration • Registration through LinkedIn OAuth
FR-2	User Confirmation	• Confirmation via Verification Email link • Confirmation via OTP (One-Time Password) SMS
FR-3	Product Catalog & Search	• Category Filtering (Mobiles, Electronics, Sports-Equipment, Fashion, Groceries) • Gender Filtering (Men, Women, Unisex) • Search Bar (Matching keywords in Title and Description) • Sorting (Popular, Price: Low to High, Price: High to Low, Discount)
FR-4	Product Details Display	• Multi-angle Product Image Carousel • Variant / Size Selection Picker • Pricing box displaying MRP, discount percentages, and final calculated price • Customer Rating and Review log display
FR-5	Shopping Cart	• Add Product with selected size and quantity • Persistent Database Sync (Restoring cart items on login) • Increment / Decrement quantity counts dynamically • Delete item from cart

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-6	Checkout & Invoice	• Shipping Address Details Form (Name, Email, Mobile, Address, Pincode) • Billing calculation (MRP, Discounts, ₹99 Delivery fee for orders < ₹999, 18% GST) • Simulated Payments (UPI, Cards, Cash on Delivery) • Order Notes / Specific Requirements configuration
FR-7	Profile & Order Tracking	• Active User Details Card (Username, Email, Account Type) • Detailed Transaction History List • Order Status Tracking (placed, packed, shipped, out for delivery, delivered, cancelled) • Self-service active order cancellation
FR-8	Admin Sales Dashboard	• Total Revenue metric aggregation • Active registered user count indicator • Listed products and total orders count summary • Top 5 recent orders feed
FR-9	Admin Inventory CRUD	•

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	• The application features an intuitive, glassmorphic user interface styled with pure CSS variables. • Includes a seamless light/dark mode theme toggle that persists in the user's localStorage across browsing sessions. • Supports fluid, responsive layouts optimized for mobile, tablet, and widescreen monitors with responsive grids.
NFR-2	Security	• User credentials are protected using bcryptjs password hashing (salt round factor of 10) before MongoDB storage. • Authenticated API requests use JWT (JSON Web Tokens) Bearer authentication

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		with a 7-day expiration. • Route endpoints are secured via Express middleware guards (protect and isAdmin) to prevent unauthorized access to administrative or user-specific data.
NFR-3	Reliability	• The backend integrates a centralized errorHandler.js that catches unhandled route exceptions and prevents the Node.js server process from crashing. • Connects to MongoDB with automatic retry configurations and exit signals (process.exit(1)) to handle database connection drops safely. • Includes self-seeding product catalog safeguards to recover inventory data if collections drop below critical levels.
NFR-4	Performance	• Frontend assets are bundled using Vite to achieve fast startup compile times and sub-second initial page loads. • Database queries are optimized with Mongoose ODM models to handle catalog fetches, search inputs, and filter checks under 100ms. • Asynchronous JavaScript APIs (e.g. Promise.all in admin dashboards) prevent blocking the server's single-threaded event loop.
NFR-5	Availability	• Decoupled client-server architecture allows the frontend static React pages to remain accessible even if the backend server goes offline (displaying clear connection warning banners instead of browser crashes). • Fully compatible with PM2 process managers or Docker containers for automated system restarts and zero-downtime deployments.
NFR-6	Scalability	• The stateless Express API server is designed to scale horizontally behind a load balancer (such as Nginx) to handle increased traffic. • The React client code is ready to be hosted on Global CDNs (like Vercel, Netlify, or AWS CloudFront) to optimize distribution speeds. • MongoDB supports read/write throughput scaling through indexing, replica sets, and database sharding.